

Project 1: Business Management System (BMS)

CONSOLE is growing fast, but our current way of working is lagging behind. Using messy spreadsheets and scattered chat messages for a firm with 150+ developers is no longer enough. We need a "Centralized Source of Truth"—one platform where everything happens.

This project is about building that professional system to stop the "digital mess" and make sure everyone is accountable.

Goal:

Your goal is to develop a **web application** that contains features solving the below mentioned problems faced by us due to lack of such centralized business management platforms. Try to make it general so that various firms can use it.

For reference you can explore platforms like [Jira | Issue & Project Tracking Software | Atlassian](#).

The 4-Stage Solution

Stage 1: The Foundation

Goal: Make a functional website.

1. Smart Login System:

- Build a secure login where the software knows your designation.
- Every single person should see completely different dashboards based on their designation.
- Face recognition for login and authorization purposes. Look up *Open-CV*.

2. Task Assignment and Monitoring:

- The employer can assign the task to its employees and can monitor them.
- While the employees get their tasks in their tasks list and can update tasks status:
 - i. To-Do → In-Progress → Completed
- *Search the term "Kanban" online, you might find something interesting*

3. The "Black Box" Recorder:

- Create a history log that records every action.
- If a task is deleted or changed, the system must show who did it and when, so there is no "mystery" behind missed deadlines.

4. Clean Interface:

- Build a simple, easy-to-use layout that works on phones and PCs.

Stage 2: The Automator and Communicator

Goal: Let the code do the boring stuff.

1. Google Calendar and Google Meet Integration:

- Tasks should automatically appear on a user's Google Calendar so they never forget a deadline.
- Auto arranging the meeting based on availability (according to their google calendar) of all persons involved.
- ****Instead of depending on Google Calendar, make a local calendar and reminding system****

2. Automatic Invoices:

- The system should automatically create PDF salary slips and bills on the 1st of every month and after confirmation automatically send it to the respective person.
- No more relying on human memory for payments.
- Sends mails regarding the bills, salary and imported announcement automatically. This time through Gmail.

3. Local Messaging or Mailing: A localized mailing or messaging for faster interaction and a centralized place for all messages/mails of tasks assigned or announcements for the employees.

Stage 3: The Live Nerve Center

Goal: Everything updates instantly.

1. Live Scoreboard:

- Build a dashboard that updates in real-time.
- As soon as a task is finished or a payment is made, the screen should update for everyone without needing to refresh.

2. Active Users:

- Show who is currently online or working on a task, just like a "live" office.

Bonus: God Mode

1. **The "Burn Rate" Graph:** Create a visual chart that predicts when the company will run out of cash based on spending. *In this don't use already existing LLMs for predictions, use a prebuilt model and integrate it to the website*
2. **Optimization:** Minimize the delays.